CS 341 - Computer Networks Quiz 1 Solutions

SETA

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Q1) Is it possible for an organization's Web server and mail server to have exactly the same alias for a hostname (for example, foo.com)? What would be the type for the RR that contains the hostname of the mail server? (2 marks)

Sol: Yes, it is possible for an organization's web server and mail server to have exactly the same alias (hostname) like foo.com. The type for the Resource Record (RR) that contains the hostname of the mail server would be **MX (Mail Exchange)**.

Q2) TCP is said to be self-clocking. Why? Justify your answer.

Sol: TCP is said to be self-clocking because it adjusts its data flow rate according to the network's capacity. This happens due to the ACKs received from the receiver, which naturally paces the transmission of new data segments. As packets are acknowledged, new packets are sent, thus making TCP self-regulating and self-clocked.

Q3)A web page consists of a base HTML file and 5 images, each requiring a separate HTTP request. System uses non-persistent HTTP and each request has an RTT of 60 ms, and the server's transmission time for each object is 15 ms, what is the total time to download the entire web page? (3 marks)

1. Given:

- 1 base HTML file + 5 images = 6 objects
- RTT = 60 ms
- Transmission time per object = 15 ms
- Non-persistent HTTP (each request requires a separate RTT) Total time:
- Base HTML file: 2 RTT + transmission time = 120 ms + 15 ms = 135 ms

- 5 images: Each requires 2 RTT + transmission time = 5 × (120 *ms* + 15 *ms*) = 675 *ms* Total time to download the entire web page = 135 *ms* + 675 *ms* = **810 ms**.

Q4) A DNS resolver needs to contact an authoritative DNS server 3 network hops away, where each hop introduces a delay of 10 ms. The resolver begins processing the DNS query immediately after the query is sent, and processing takes 5 ms. Meanwhile, the server responds in 50 ms. The resolver can process and receive the response simultaneously. Calculate the total time taken for the DNS query, assuming processing overlaps with propagation. (3 marks)

Solution:

Time for the DNS query = Time for DNS propagation and server response (since processing overlaps)

Total time = (2 × Propagation time) + Server response time

= 2 × (3 × 10 ms) + 50 ms = 110 ms. NB: No need to calculate the transmission time

Q5) An email of size 1 MB is sent using SMTP over a network with a data rate of 2 Mbps. The initial connection setup takes 3 RTTs, each RTT being 40 ms. However, after the first 500 KB of the email is sent, the server processes it while the client continues to send the remaining data. Calculate the total time to send the email, considering the server begins processing halfway through the transmission. (3 marks)

Solution:

Given:

- Email size = 1 MB = 8,192,000 bits
- Data rate = 2 Mbps
- RTT = 40 ms
- Initial connection setup = 3 RTTs = 3 × 40 ms = 120 ms
- Server begins processing after 500 KB (4,096,000 bits) Total time:
- Time to send 500 KB = $\frac{4{,}096{,}000 bits}{2Mbps}$ = 2.048 s
- Time to send remaining 500 KB = 2.048 s (simultaneous with server pro- cessing)
- Total time = 120 $ms(connectionsetup)$+2.048 $s(firsthalf)$+2.048 $s(secondhalf)$ = 4216 ms

Q6) During TCP slow start, if the initial congestion window is 1 MSS and doubles every RTT, how large will the congestion window be after 4 RTTs? The RTT increases by 10 ms each RTT starting from an initial RTT of 50 ms, and this increase affects the doubling rate.

Given:

- Initial congestion window = 1 MSS
- Window doubles every RTT
- RTT increases by 10 ms each RTT starting

from 50 ms Congestion window after 4 RTTs:

- 1st RTT: $1MSS \times 2 = 2MSS$ (RTT = 50 ms)
- 2nd RTT: $2MSS \times 2 = 4MSS$ (RTT = 60 ms)
- 3rd RTT: $4MSS \times 2 = 8MSS$ (RTT = 70 ms)
- 4th RTT: $8MSS \times 2 = 16MSS$ (RTT = 80 ms)

Final congestion window after 4 RTTs = **16 MSS**.

Q7) Hosts A and B are communicating over a TCP connection, and Host B has already received from A all bytes up through byte 126. Suppose Host A then sends two segments to Host B back-to-back. The first and second segments contain 80 and 40 bytes of data, respectively. In the first segment, the sequence number is 127, the source port number is 302, and the destination port number is 80. Host B sends an acknowledgement whenever it receives a segment from Host A.
(0.5+0.5+1+2 marks)
Given:

- Sequence number of first segment = 127
- Source port number = 302
- Destination port number = 80
- First segment size = 80 bytes
- Second segment size = 40 bytes

a) In the second segment sent from Host A to B, what are the sequence number, source port number, and destination port number?

(a) Second Segment:

- Sequence number = 127 + 80 = **207**
- Source port number = **302**
- Destination port number = **80**

b) If the first segment arrives before the second segment, in the acknowledgement of the first arriving segment, what is the acknowledgement number, the source port number, and the destination port number?

(b) Acknowledgment of First Segment:

- Acknowledgment number = 127 + 80 = **207**
- Source port number = **80**
- Destination port number = **302**

c) If the second segment arrives before the first segment, in the acknowledgement of the first arriving segment, what is the acknowledgement number?

(c) Acknowledgment if Second Segment Arrives First:

- Acknowledgment number = **127** (since it expects the first segment).

d) Suppose the two segments sent by A arrive in order at B. The first acknowledgement arrives after the first timeout interval. Draw the timing diagram, showing these segments and all other segments and acknowledgements sent. Assume that there is no additional packet loss. For each segment in your figure, provide the sequence number and number of bytes of data; for each acknowledgement that you add, provide the acknowledgement number.

(d) Timing Diagram (Explanation):

- 1st segment sent with sequence number 127 (80 bytes)
- 2nd segment sent with sequence number 207 (40 bytes)
- 1st ACK sent after receiving the first segment, ACK number 207
- 2nd ACK sent after receiving the second segment, ACK number 247
- ACK numbers correspond to the next expected byte.